THE OLD MUTUAL SOUTH AFRICAN MATHEMATICS OLYMPIAD

Organised by the
SOUTH AFRICAN MATHEMATICS FOUNDATION

2025 SECOND ROUND
SENIOR SECTION:
GRADE 10 - 12

15 May 2025 **Time: 120 minutes** **Number of questions: 25**

Instructions

1. *The answers to all questions are integers from 0 to 999. Each question has only one correct answer.*
2. *Scoring rules:*
   2.1. *Each correct answer is worth 3 marks in Part A, 4 marks in Part B and 6 marks in Part C.*
   2.2. *There is no penalty for an incorrect answer or any unanswered question.*
3. *You must use an HB pencil. Rough work paper, a ruler and an eraser are permitted.* ***Calculators and geometry instruments are not permitted.***
4. *Figures are not necessarily drawn to scale.*
5. *Indicate your answers on the sheet provided.*
6. *Start when the invigilator tells you to do so.*
7. *Answers and solutions will be available at* `www.samf.ac.za`

Do not turn the page until you are told to do so.
Draai die boekie om vir die Afrikaanse vraestel.

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Organisations involved: AMESA, SA Mathematical Society,
SA Akademie vir Wetenskap en Kuns

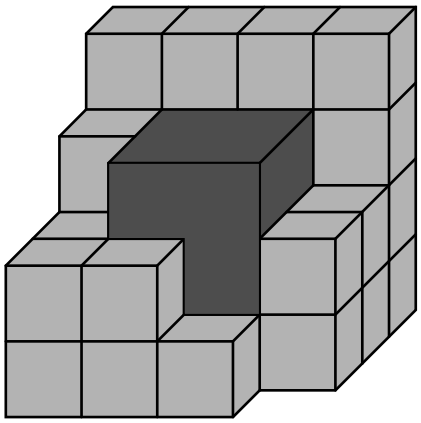
SECTION A (3 MARKS PER QUESTION)

1) Find the value of $(2025)^2 - 2(2025)(2020) + (2020)^2$.

2) If $2^x = 8$, $8^y = 16$ and $16^z = 32$, determine xyz.

3) Five apples and five bananas are worth the same as two bananas and eleven apples. How many apples is a banana worth?

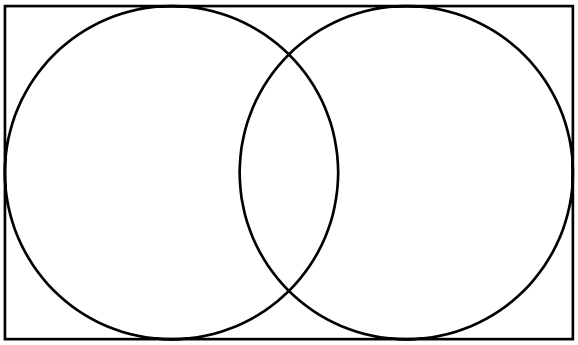
4) The figure shows a dark shaded cube partially enclosed by smaller light shaded cubes. There are no cubes missing on the faces not visible in the picture. How many extra light shaded cubes are needed to enclose the dark shaded cube and complete the large four by four cube?



5) Lisa's bucket weighs 21 kg when full of water. After she pours out half the water from the bucket, the bucket and the remaining water weighs 12 kg. What is the weight in kilograms of the empty bucket?

6) The sum of the digits of a two-digit number is 7. If a smaller new number is created by swapping the digits of the original number, the product of the original number and the new number is 976. Determine the original number.

7) Two circles are enclosed in a rectangle measuring 9 by 5. What is the distance between the centres of the circles?



8) I hiked into the forest at 4 km/h and returned by the same route at 3 km/h. If the round trip took $3\frac{1}{2}$ hours, how many kilometres into the forest did I hike?

9) Two tractors are ploughing a three-hectare field. The older tractor takes 10 hours to plough 1 hectare, and the newer tractor takes 5 hours to plough 1 hectare. Both tractors work together for 2 hours, but then the older tractor breaks down. How many hours is it going to take the newer tractor to complete the ploughing of the three-hectare field?

10) In how many ways can six students, A, B, C, D, E and F, sit together in the first row during assembly if students A and B must sit next to each other and students C and D must sit next to each other?

11) A trapezium has area 300 and parallel sides a and b. If the height h between the two parallel sides is increased by 10%, and the lengths of both sides a and b are decreased by 10%, determine the area of the new trapezium.

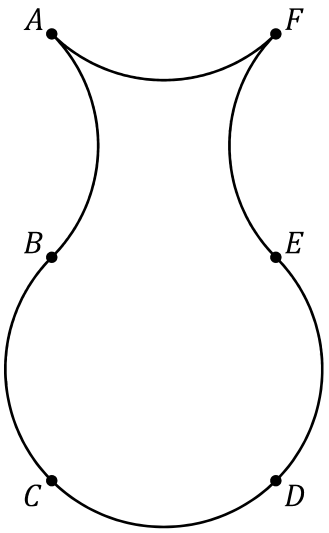
12) Find the value of x if

$$x = \sqrt{6 - \sqrt{6 - \sqrt{6 - \sqrt{6 - \sqrt{6 - \cdots}}}}}$$

13) The probability that at least two of the three children in a family are girls is $\frac{x}{56}$. The probability to have a girl or a boy each equal a half. Determine the value of x.

14) Four students each chooses a number. When each student's number is added to twice the mean of the other three numbers, the following sums are obtained: 42, 54, 66 and 78. Determine the mean of the four numbers chosen by the students.

15) The vase-like shape in the sketch is formed by six circular arcs, AB, BC, CD, DE, EF and FA, each which is a quarter of the circumference of a circle with radius 6, all joined together. What is the area of the vase?



SECTION B (5 MARKS PER QUESTION)

16) Contestants in a singing competition must each select two songs from a list of 20 to perform. What is the least number of contestants that need to enter the competition to be sure that at least two contestants make the same selection?

17) Suppose

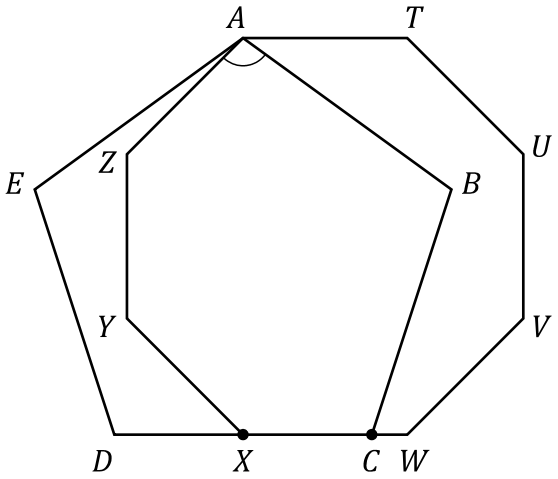
$$\begin{aligned} a+b+c+d+e &= 20 \\ a+b+c+d+f &= 19 \\ a+b+c+e+f &= 18 \\ a+b+d+e+f &= 17 \\ a+c+d+e+f &= 16 \\ b+c+d+e+f &= 15 \end{aligned}$$

Determine $a \times b \times c \times d \times e \times f$.

18) The sum of 50 different numbers from the set $\{1; 2; \ldots; 100\}$ is equal to 2900. What is the smallest number of even numbers that could be part of this sum?

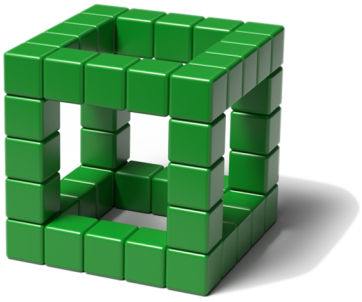
19) Find the smallest positive integer n so that each digit of $15 \times n$ is a 0 or an 8.

20) In the figure $ABCDE$ is a regular pentagon and $ATUVWXYZ$ is a regular octagon with point X on CD and point C on WX. Determine the size of $Z\hat{A}B$ in degrees.

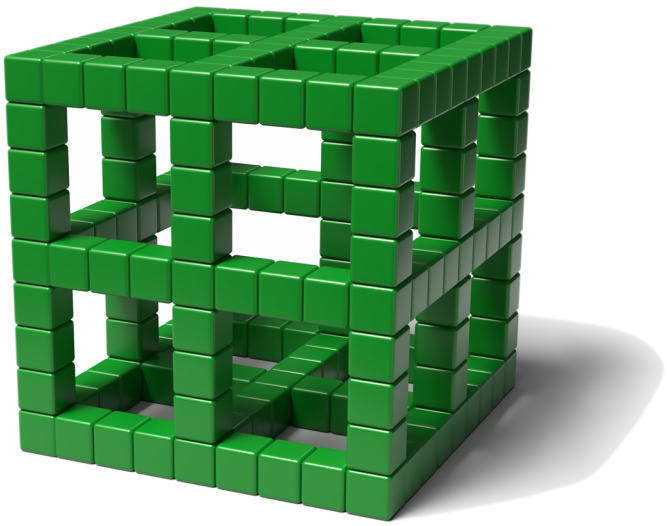


SECTION C (6 MARKS PER QUESTION)

21) The first two elements in a sequence of hollow cubes, constructed from small cubes according to a pattern, are shown in the 3D-figures. The six faces of each element are identical. If the pattern continues, what is the total number of 3 by 3 holes that the element constructed from exactly 674 small cubes will have on its faces?



Element 1



Element 2

22) Point P is 9 units from the centre of a circle with radius 15. How many chords of the circle, including the diameter, pass through P and have integer lengths?

23) If a, b and c are distinct positive integers such that

$$abc + ab + bc + ac + a + b + c = 1000,$$

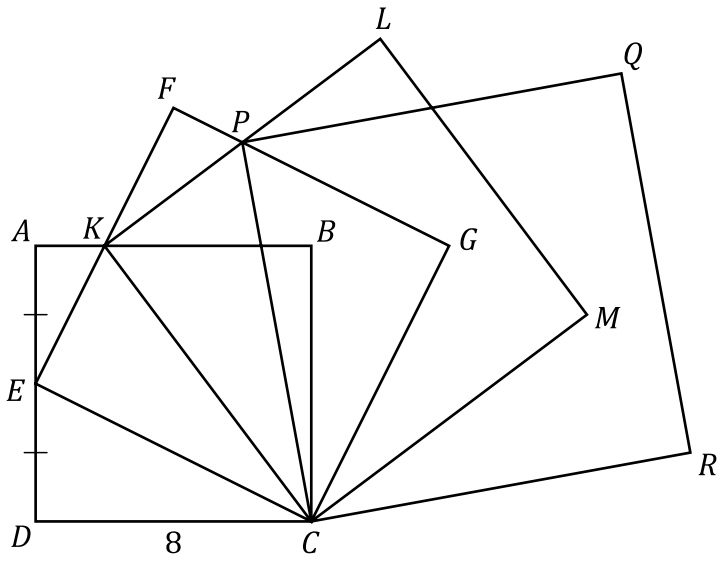
find the value of $a + b + c$.

24) If f is defined on the set of integers and satisfies

$$f(n) = \begin{cases} n - 3, & \text{if } n \geq 1000 \\ f\big(f(n+5)\big), & \text{if } n < 1000 \end{cases}$$

determine $f(25)$.

25) The squares $ABCD$, $EFGC$, $KLMC$ and $PQRC$ are shown in the figure, with point E on AD, point K the intersection of AB and EF, and point P the intersection of FG and KL. The length of CD is 8 and $AE = DE$. What is the value of FL^2?



22) Punt P is 9 eenhede vanaf die middelpunt van 'n sirkel met radius 15. Hoeveel koorde van die sirkel, insluitend die middellyn, gaan deur P en het heeltallige lengtes?

23) Indien a, b en c positiewe heelgetalle is, sodanig dat

$$abc + ab + bc + ac + a + b + c = 1000,$$

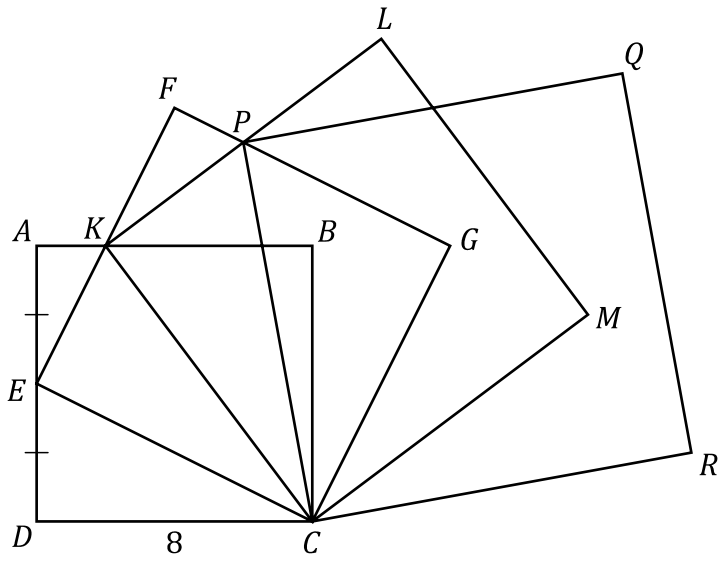
vind die waarde van $a + b + c$.

24) As f gedefinieer is vir die versameling heelgetalle en

$$f(n) = \begin{cases} n - 3, & \text{if } n \geq 1000 \\ f\big(f(n+5)\big), & \text{if } n < 1000 \end{cases}$$

bevredig, bepaal $f(25)$.

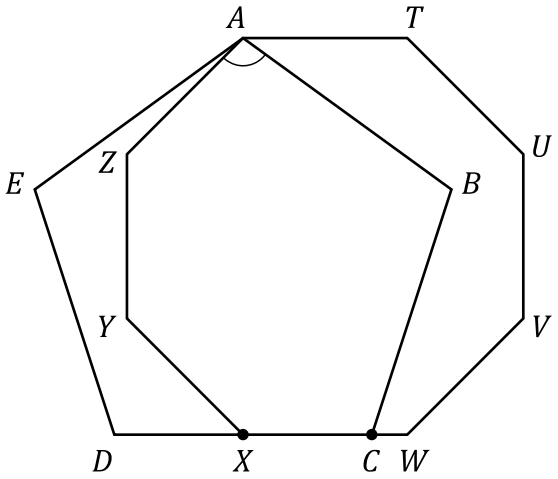
25) Die vierkante $ABCD$, $EFGC$, $KLMC$ en $PQRC$ word in die figuur getoon, met punt E op AD, K die snypunt van AB en EF, en P die snypunt van FG en KL. Die lengte van CD is 8 en $AE = DE$. Wat is die waarde van FL^2?



18) Die som van 50 verskillende getalle in die versameling $\{1; 2; \ldots; 100\}$ is gelyk aan 2900. Wat is kleinste aantal ewe getalle wat deel van hierdie som kan wees?

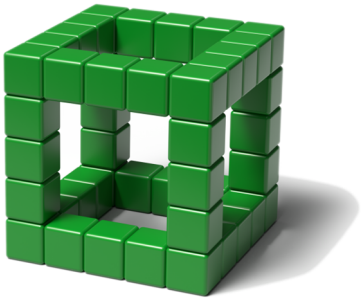
19) Vind die kleinste positiewe heelgetal n sodat elke syfer van $15 \times n$ óf 'n 0 óf 'n 8 is.

20) In die skets is $ABCDE$ 'n reëlmatige vyfhoek en $ATUVWXYZ$ 'n reëlmatige agthoek, met punt X op CD en punt C op WX. Bepaal die grootte van $Z\hat{A}B$ in grade.

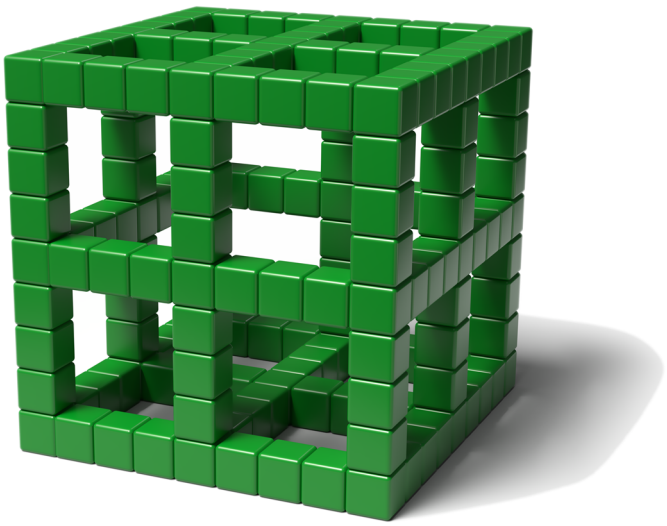


AFDELING C (6 PUNTE PER VRAAG)

21) Die eerste twee elemente in 'n ry hol kubusse, wat volgens 'n patroon met klein kubusse gevorm word, word in die 3D-figure getoon. Die ses vlakke van elke element is identies. Indien die patroon voortgesit word, wat is die totale aantal 3 by 3 gate wat die element wat met presies 674 klein kubusse gevorm word, op al sy vlakke saam sal hê?



Element 1

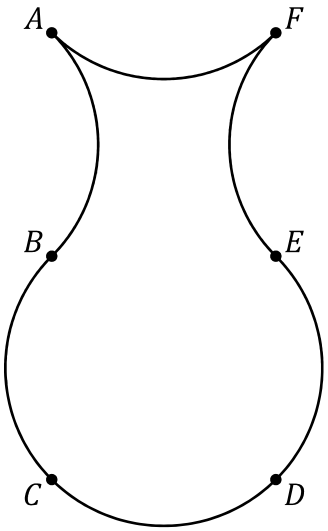


Element 2

13) Die waarskynlikheid dat ten minste twee van die drie kinders in 'n gesin dogters is, is $\frac{x}{56}$. Die waarskynlikheid om 'n seun of 'n dogter te hê is beide gelyk aan 'n half. Bepaal die waarde van x.

14) Vier studente kies elkeen 'n getal. Wanneer elke student se getal by twee keer die gemiddeld van die ander drie getalle getel word, word die volgende resultate verkry: 42, 54, 66 en 78. Bepaal die gemiddeld van die vier getalle wat deur die studente gekies is.

15) Die vaas in die skets word gevorm deur ses sirkelboë, AB, BC, CD, DE, EF en FA, wat elkeen 'n kwart van die omtrek van 'n sirkel met radius 6 is, aan mekaar te voeg. Wat is die oppervlakte van die vaas?



AFDELING B (5 PUNTE PER VRAAG)

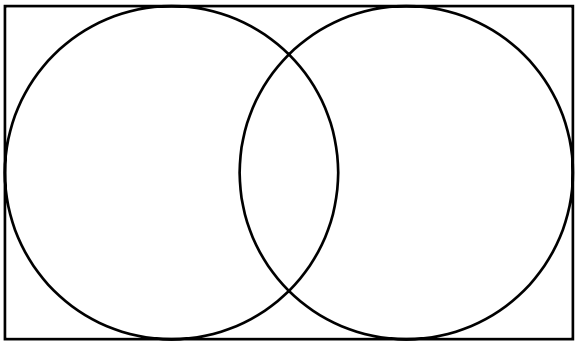
16) Deelnemers van 'n sangkompetisie moet elkeen twee liedjies uit 'n lys van 20 kies. Wat is die kleinste getal deelnemers wat vir die kompetisie moet inskryf sodat daar ten minste twee deelnemers is wat dieselfde keuse van liedjies gemaak het?

17) Gestel

$$\begin{aligned} a+b+c+d+e &= 20 \\ a+b+c+d+f &= 19 \\ a+b+c+e+f &= 18 \\ a+b+d+e+f &= 17 \\ a+c+d+e+f &= 16 \\ b+c+d+e+f &= 15 \end{aligned}$$

Bepaal $a \times b \times c \times d \times e \times f$.

7) Twee sirkels is ingeslote binne ’n reghoek met mates van 9 by 5. Wat is die afstand tussen die middelpunte van die sirkels?



8) Ek stap in ’n woud in teen 4 km/h en keer terug op dieselfde roete teen 3 km/h. As die staptog my $3\frac{1}{2}$ uur neem, hoeveel kilometer het ek in die woud in gestap?

9) Twee trekkers word gebruik om ’n drie hektaar veld te ploeg. Die ouer trekker neem 10 ure om 1 hektaar te ploeg, terwyl die nuwer trekker 5 ure neem om 1 hektaar te ploeg. Albei trekkers werk gelyktydig vir 2 ure, maar dan breek die ouer trekker. Hoeveel ure sal dit die nuwer trekker neem om die drie hektaar veld klaar te ploeg?

10) Op hoeveel maniere kan ses studente, A, B, C, D, E en F, bymekaar in dit voorste ry van die saal sit as studente A en B langs mekaar moet sit en studente C en D langs mekaar moet sit?

11) ’n Trapesium het ’n oppervlakte van 300 en ewewydige sye a en b. Indien die hoogte h tussen die twee parallelle sye met 10% vergroot word, terwyl die lengtes van beide sye a en b met 10% verklein word, bepaal die oppervlakte van die nuwe trapesium.

12) Vind die waarde van x indien

$$x = \sqrt{6 - \sqrt{6 - \sqrt{6 - \sqrt{6 - \sqrt{6 - \cdots}}}}}$$

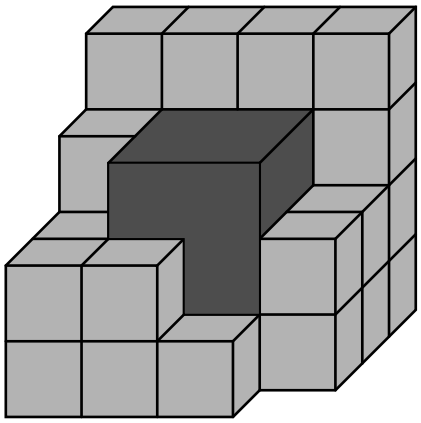
AFDELING A (3 PUNTE PER VRAAG)

1) Vind die waarde van $(2025)^2 - 2(2025)(2020) + (2020)^2$.

2) As $2^x = 8$, $8^y = 16$ en $16^z = 32$, bepaal xyz.

3) Vyf appels en vyf piesangs het dieselfde waarde as twee piesangs en elf appels. Hoeveel appels is 'n piesang werd?

4) In die figuur word 'n donker ingekleurde kubus getoon wat gedeeltelik deur kleiner lig ingekleurde kubusse bedek word. Daar is geen kubusse uitgelaat op die kante wat nie in die skets gesien kan word nie. Hoeveel ekstra lig ingekleurde kubusse word benodig om die donker ingekleurde kubus ten volle te bedek en die groot vier by vier kubus te voltooi?



5) Lisa se emmer weeg 21 kg as dit vol water is. Nadat sy die helfte van die water in die emmer uitgegooi het, weeg die emmer en die oorblywende water 12 kg. Wat is die gewig van die leë emmer, in kilogram?

6) Die som van die syfers van 'n twee-syfergetal is 7. As 'n nuwe, kleiner getal gevorm word deur die syfers van die oorspronklike getal om te ruil, is die produk van die oorspronklike getal en die nuwe getal 976. Bepaal die oorspronklike getal.

DIE OLD MUTUAL SUID-AFRIKAANSE WISKUNDE-OLIMPIADE

Georganiseer deur die
SOUTH AFRICAN MATHEMATICS FOUNDATION

2025 TWEEDE RONDTE
SENIOR AFDELING:
GRAAD 10 - 12

15 Mei 2025 **Tyd: 120 minute** **Aantal vrae: 25**

Instruksies

1. *Die antwoorde op al die vrae is heelgetalle van 0 tot 999. Elke vraag het slegs een korrekte antwoord.*
2. *Puntetoekenning:*
   2.1. Elke korrekte antwoord tel 3 punte in Afdeling A, 4 punte in Afdeling B en 6 punte in Afdeling C.
   2.2. Geen punte word afgetrek vir foutiewe antwoorde of onbeantwoorde vrae nie.
3. *Gebruik 'n HB potlood. Papier vir rofwerk, 'n liniaal en uitveër word toegelaat.* ***Sakrekenaars en meetkunde-instrumente word nie toegelaat nie.***
4. *Figure is nie noodwendig volgens skaal geteken nie.*
5. *Beantwoord die vrae op die antwoordblad wat voorsien word.*
6. *Begin sodra die toesighouer die teken gee.*
7. *Antwoorde en oplossings sal beskikbaar wees by* `www.samf.ac.za`

Moenie omblaai voordat dit aan jou gesê word nie.
Turn the booklet over for the English paper.

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